Nonhomogeneous Equations & Undetermined Coefficients

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: General Solution Structure**

Find the general real solution of $\mathbf{x}' = (-2,\ -5;\ 1,\ 0)\mathbf{x}$, classify the origin, and determine whether trajectories rotate clockwise or counterclockwise.

Step 1: Characteristic polynomial: $\det(-2-\lambda,\ -5;\ 1,\ -\lambda) = \lambda(\lambda+2)+5 = \lambda^2+2\lambda+5 = 0 \Rightarrow \lambda = -1 \pm 2i.$ Since $\alpha = -1 < 0$, the origin is an asymptotically stable spiral sink. Step 2: Eigenvector for $\lambda = -1+2i$:

$(-1-2i,\ -5;\ 1,\ 1-2i)(v_1;\ v_2) = \mathbf{0} \Rightarrow v_1 + (1-2i)v_2 = 0 \Rightarrow \mathbf{v} = (-1+2i;\ 1) = (-1;\ 1) + i(2;\ 0).$ Step 3: Real and i.

2 **[Foundational]** • **Target: Exponential and Sinusoidal Forcing Functions**

Consider the differential equation $y'' - 3y' + 2y = f(t)$ with initial conditions $y(0) = 0$ and $y'(0) = 0$. If the forcing function is $f(t) = u(t-2)$, where $u(t)$ is the Heaviside step function, what is the Laplace transform of the solution $Y(s) = \mathcal{L}\{y(t)\}$?

The Laplace transform of the differential equation is $s^2Y(s) - sy(0) - y'(0) - 3(sY(s) - y(0)) + 2Y(s) = \mathcal{L}\{f(t)\}$. Given $y(0) = 0$ and $y'(0) = 0$, this simplifies to $s^2Y(s) - 3sY(s) + 2Y(s) = \mathcal{L}\{u(t-2)\}$, so $Y(s)(s^2 - 3s + 2) = \mathcal{L}\{u(t-2)\}$. The Laplace transform of $u(t-a)$ is $\frac{e^{-as}}{s}$. Therefore, $\mathcal{L}\{u(t-2)\} = \frac{e^{-2s}}{s}$. Thus, $Y(s) = \frac{e^{-2s}}{s(s^2-3s+2)}$. Factoring the denominator, $s^2 - 3s + 2 = (s-1)(s-2)$. So.

3 **[Foundational]** • **Target: The Modification Rule (Resonance / Overlap)**

Solve the double-resonance equation $y'' - 6y' + 9y = 4e^{3t}$ with $y(0) = 1$ and $y'(0) = 3$.

Step 1: Homogeneous: $r^2 - 6r + 9 = (r-3)^2 = 0 \Rightarrow y_c(t) = (C_1 + C_2t)e^{3t}$. Step 2: Resonance multiplication: Both e^{3t} and te^{3t} are in y_c (multiplicity $s = 2$). Multiply trial by t^2: $y_p(t) = At^2e^{3t}$. Using operator $(D-3)^2[At^2e^{3t}] = e^{3t}D^2[At^2] = e^{3t}(2A) = 4e^{3t} \Rightarrow 2A = 4 \Rightarrow A = 2$. Thus $y_p(t) = 2t^2e^{3t}$. Step 3: Initial conditions: $y(t) = (C_1 + C_2t + 2t^2)e^{3t}$. $y(0) = C_1 = 1$..

4 **[Foundational]** • **Target: The Principle of Superposition**

Construct the Green's kernel $G(x,t)$ for $L[y] = y'' - y$, and express the particular solution of $y'' - y = \frac{1}{e^x+1}$ satisfying $y(0) = 0, y'(0) = 0$ as an explicit definite integral.

Step 1: Basis & Wronskian: $y_1(x) = e^x$, $y_2(x) = e^{-x}$. $W(t) = e^t(-e^{-t}) - e^t(e^{-t}) = -2$. Step 2: Green's kernel: $G(x,t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W(t)} = \frac{e^te^{-x} - e^xe^{-t}}{-2} = \frac{e^{x-t} - e^{-(x-t)}}{2} = \sinh(x-t).$ Step 3: Integral representation: $y(x) = \int_0^x \sinh(x-t) \cdot \frac{1}{e^t+1}\,dt.$ Verifying: $G(t,t) = \sinh(0) = 0$, and $\frac{\partial G}{\partial x}\big|_{x=t} = \cosh(0) = 1$, satisfying the universal Cauchy impulse propert.

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5 [Foundational] • Target: General Solution Structure

Find a particular solution of $y'' + 4y = \sec(2x)$ for $x \in (-\pi/4, \pi/4)$ using Variation of Parameters.

Step 1: Basis & Wronskian: $y_1 = \cos 2x,\ y_2 = \sin 2x$. $W = \cos 2x(2\cos 2x) - (-2\sin 2x)\sin 2x = 2$. Step 2: Parameter integrals: $u_1'(x) = -\frac{y_2(x)g(x)}{W} = -\frac{\sin 2x \sec 2x}{2} = -\frac{1}{2}\tan 2x \Rightarrow u_1(x) = \frac{1}{4}\ln|\cos 2x|$. $u_2'(x) = \frac{y_1(x)g(x)}{W} = \frac{\cos 2x \sec 2x}{2} = \frac{1}{2} \Rightarrow u_2(x) = \frac{x}{2}$. Step 3: Particular solution: $y_p(x) = u_1 y_1 + u_2 y_2 = \frac{1}{4}\cos(2x)\ln|\cos 2x| + \frac{x}{2}\sin(2x)$.

6 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

Find a particular solution of $y'' + 2y' + y = \frac{e^{-x}}{x}$ for $x > 0$ using Variation of Parameters.

Step 1: Basis & Wronskian: $y_1 = e^{-x},\ y_2 = xe^{-x}$. $W = e^{-x}(e^{-x} - xe^{-x}) - (-e^{-x})xe^{-x} = e^{-2x}$. Step 2: Parameter integrals: $u_1'(x) = -\frac{xe^{-x}(e^{-x}/x)}{e^{-2x}} = -1 \Rightarrow u_1(x) = -x$. $u_2'(x) = \frac{e^{-x}(e^{-x}/x)}{e^{-2x}} = \frac{1}{x} \Rightarrow u_2(x) = \ln x$. Step 3: Particular solution: $y_p(x) = (-x)e^{-x} + (\ln x)xe^{-x} = xe^{-x}(\ln x - 1)$.

7 [Foundational] • Target: The Modification Rule (Resonance / Overlap)

Solve the initial value problem with step forcing using the Laplace transform:

$y'' + 4y = 8u(t-2), \qquad y(0) = 0, \quad y'(0) = 0.$

Step 1: Transform equation: $(s^2 + 4)Y(s) = \frac{8e^{-2s}}{s} \Rightarrow Y(s) = 8e^{-2s} \cdot \frac{1}{s(s^2+4)}$. Step 2: Partial fractions on transfer factor: $\frac{1}{s(s^2+4)} = \frac{1/4}{s} - \frac{s/4}{s^2+4} \Rightarrow \mathcal{L}^{-1}\left\{\frac{8}{s(s^2+4)}\right\} = 2 - 2\cos(2t)$. Step 3: Apply Second Shift Theorem: $y(t) = u(t-2)(2 - 2\cos(2(t-2)))$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The Principle of Superposition

Consider the separable differential equation $\frac{dy}{dx} = ky$ where k is a positive constant. A solution curve passes through the points $(0, 5)$ and $(2, 20)$. Determine the value of k and then find the value of y when $x = 3$.

The differential equation $\frac{dy}{dx} = ky$ is separable. Separating variables gives $\frac{dy}{y} = k\,dx$. Integrating both sides, we get $\int \frac{dy}{y} = \int k\,dx$, which yields $\ln|y| = kx + C_1$. Exponentiating both sides gives $|y| = e^{kx + C_1} = e^{C_1}e^{kx}$. Let $C = \pm e^{C_1}$, so $y = Ce^{kx}$. We are given that the solution curve passes through $(0, 5)$. Substituting these values into $y = Ce^{kx}$: $5 = Ce^{k(0)} \Rightarrow 5 = C$. Thus, the particular solution is $y = 5e^{kx}$. We ar.

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9 **[Intermediate]** • **Target: General Solution Structure**

Consider the differential equation $\frac{dy}{dx} = \frac{x^2\cos(y)}{1-x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. If there exists a unique solution in a neighborhood of $x = 0$ and $y = \frac{\pi}{2}$, find the value of $y'(0)$.

The given differential equation is $\frac{dy}{dx} = \frac{x^2\cos(y)}{1-x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. We are asked to find $y'(0)$. The differential equation directly gives us the expression for $\frac{dy}{dx}$, which is $y'(x)$. To find $y'(0)$, we substitute $x = 0$ and $y = y(0) = \frac{\pi}{2}$ into the differential equation: $y'(0) = \frac{(0)^2\cos(y(0))}{1-(0)^3}$ Substitute $y(0) = \frac{\pi}{2}$: $y'(0) = \frac{0 \cdot \cos(\frac{\pi}{2})}{1-0}$ We know that.

10 **[Intermediate]** • **Target: Exponential and Sinusoidal Forcing Functions**

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. You are asked to approximate $y(0.2)$ using Euler's method. If the step size $h = 0.1$ is used, and it is known that the exact solution at $x = 0.2$ is $y(0.2) = 1.24$, determine the value of k such that $y(0.2)_{\text{Euler}} = 1.24 - k$.

Euler's method for approximating the solution to $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$ and step size h is given by $y_{n+1} = y_n + hf(x_n, y_n)$. In this problem, $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.1$. We want to approximate $y(0.2)$ with $h = 0.1$. This requires two steps: Step 1: From $x_0 = 0$ to $x_1 = 0.1$. $y_1 = y_0 + hf(x_0, y_0)$ $y_1 = 1 + 0.1(0 + 1)$ $y_1 = 1 + 0.1(1)$ $y_1 = 1 + 0.1$ $y_1 = 1.1$ Step 2: From $x_1 = 0.1$ to $x_2 = 0.2$..

11 **[Intermediate]** • **Target: The Modification Rule (Resonance / Overlap)**

Find a particular solution of $y'' - 10y' + 25y = e^{5x}x\ln x$ (for $x > 0$) using variation of parameters.

Homogeneous: $r^2 - 10r + 25 = 0$ has roots $r = 5$ (repeated), so $y_1 = e^{5x}$, $y_2 = xe^{5x}$. $u_1 = -\frac{x^3\log(x)}{3} + \frac{x^3}{9}$, $u_2 = \frac{x^2\log(x)}{2} - \frac{x^2}{4}$. So $y_p = \frac{x^3e^{5x}\log(x)}{6} - \frac{5x^3e^{5x}}{36}$. Final Answer: $\frac{x^3e^{5x}\log(x)}{6} - \frac{5x^3e^{5x}}{36}$

12 **[Intermediate]** • **Target: The Principle of Superposition**

Find the coefficient a_5 of the power series solution $y = \sum a_n x^n$ of $y'' - xy' - y = 0$ with $y(0) = 1$, $y'(0) = 2$.

Substitute the series and match powers of x to get a recurrence; with $a_0 = 1,\ a_1 = 2$ the coefficients are $a_0 = 1, a_1 = 2, a_2 = \frac{1}{2}, a_3 = \frac{2}{3}, a_4 = \frac{1}{8}, a_5 = \frac{2}{15}$. Final Answer: $\frac{2}{15}$

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13 **[Intermediate]** • **Target: General Solution Structure**

A harvested population evolves according to $\frac{dP}{dt} = P\left(1 - \frac{P}{10}\right) - 2.1$. Determine the two equilibrium populations, identify which is the extinction threshold, and determine whether a starting population $P(0) = 2$ survives or collapses.

Step 1: Equilibrium quadratic: Set $\frac{dP}{dt} = 0$: $-\frac{P^2}{10} + P - 2.1 = 0 \Rightarrow P^2 - 10P + 21 = 0$. Factoring: $(P-3)(P-7) = 0 \Rightarrow P_1 = 3,\ P_2 = 7$. Step 2: Stability: $f'(P) = 1 - \frac{P}{5}$. $f'(3) = 1 - 0.6 = +0.4 > 0 \Rightarrow P = 3$ is an unstable extinction threshold. $f'(7) = 1 - 1.4 = -0.4 < 0 \Rightarrow P = 7$ is the stable carrying capacity. Step 3: Dynamics for $P(0) = 2$: Since $P(0) = 2 < 3$, $\frac{dP}{dt} < 0$. The population strictly declines and collapses to.

14 **[Intermediate]** • **Target: Exponential and Sinusoidal Forcing Functions**

Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = 1$. Determine the exact time $t > 0$ at which $y(t)$ attains its absolute maximum value.

Step 1: Characteristic roots: $r^2 + 4r + 4 = (r+2)^2 = 0 \Rightarrow r = -2$ (repeated). General solution: $y(t) = (C_1 + C_2 t)e^{-2t}$. Step 2: Initial conditions: $y(0) = C_1 = 2$.
$y'(t) = C_2 e^{-2t} - 2(C_1 + C_2 t)e^{-2t} = ((C_2 - 2C_1) - 2C_2 t)e^{-2t}$. $y'(0) = C_2 - 2(2) = 1 \Rightarrow C_2 = 5$. Particular solution: $y(t) = (2 + 5t)e^{-2t}$. Step 3: Find maximum: Set $y'(t) = 0$:
$((5-4) - 10t)e^{-2t} = (1 - 10t)e^{-2t} = 0 \Rightarrow 1 - 10t = 0 \Rightarrow t_{\max} = \frac{1}{10} = 0.1$. Maximu.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The Modification Rule (Resonance / Overlap)** • **Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Consider the stiff decay equation $\frac{dy}{dt} = -100y$ with $y(0) = 1$. Determine the stability threshold h_{crit} for Forward Euler, and compute the value of $y(0.1)$ using Backward (Implicit) Euler with step size $h = 0.05$.

Step 1: Forward Euler stability threshold: For $y' = \lambda y$ with $\lambda = -100$, the amplification factor is $1 + h\lambda = 1 - 100h$. Stability requires: $|1 - 100h| < 1 \Rightarrow -1 < 1 - 100h < 1 \Rightarrow h < \frac{2}{100} = 0.02$. Thus $h_{\text{crit}} = 0.02$. If $h > 0.02$, Forward Euler explodes uncontrollably. Step 2: Backward (Implicit) Euler formula:
$y_{n+1} = y_n + h(-100y_{n+1}) \Rightarrow (1 + 100h)y_{n+1} = y_n \Rightarrow y_{n+1} = \frac{1}{1+100h} y_n$. With $h = 0.05$.

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16 **[Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

Consider the differential equation $\frac{dy}{dx} = \frac{\cos(x)}{y^2}$ for $y > 0$ and $x \in [0, \frac{\pi}{2}]$. If $y(0) = 2$, find the value of $y(\frac{\pi}{2})$.

The given differential equation is $\frac{dy}{dx} = \frac{\cos(x)}{y^2}$. This is a separable differential equation. We can rewrite it as $y^2 dy = \cos(x) dx$. Integrating both sides, we get: $\int y^2 dy = \int \cos(x) dx$ $\frac{y^3}{3} = \sin(x) + C$ We are given the initial condition $y(0) = 2$. Substituting these values into the integrated equation: $\frac{2^3}{3} = \sin(0) + C$ $\frac{8}{3} = 0 + C$ $C = \frac{8}{3}$ So, the particular solution is: $\frac{y^3}{3} = \sin(x) + \frac{8}{3}$ We need to find the.

17 **[Challenging] • Target: General Solution Structure • Cross-Lesson: 5.01 Variation of Parameters**

Use the Laplace transform and the Convolution Theorem to solve the integral equation $y(t) = t^2 + \int_0^t e^{t-\tau} y(\tau)\, d\tau$.

Step 1: Recognize convolution: The equation is $y(t) = t^2 + e^t * y(t)$. Step 2: Transform both sides: $Y(s) = \frac{2}{s^3} + (\frac{1}{s-1})Y(s) \Rightarrow Y(s)(1 - \frac{1}{s-1}) = \frac{2}{s^3}$. $Y(s)(\frac{s-2}{s-1}) = \frac{2}{s^3} \Rightarrow Y(s) = \frac{2(s-1)}{s^3(s-2)}$. Step 3: Partial fractions: $\frac{2(s-1)}{s^3(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s-2}$. By cover-up, $D = \frac{2(2-1)}{2^3} = \frac{2}{8} = \frac{1}{4}$.

18 **[Challenging] • Target: Exponential and Sinusoidal Forcing Functions • Cross-Lesson: 1.01 Separable Equations**

Consider the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Use the Euler method with step size $h = 0.5$ to approximate $y(1)$.

The Euler method approximates the solution of an ODE $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$ using the iterative formula $y_{i+1} = y_i + hf(x_i, y_i)$, where h is the step size and $x_i = x_0 + ih$. Given: $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.5$. We want to approximate $y(1)$. Since $1 = 0 + 2 \times 0.5$, we need two steps. Step 1: Approximate $y(0.5)$. $x_1 = x_0 + h = 0 + 0.5 = 0.5$

$y_1 = y_0 + hf(x_0, y_0) = 1 + 0.5(0 + 1) = 1 + 0.5(1) = 1.5$. Step 2: Approximate.

19 **[Challenging] • Target: The Modification Rule (Resonance / Overlap) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Solve $\frac{dy}{dx} = \frac{1+y^2}{x^2-4}$ on $(-2, 2)$ with $y(0) = 1$. At what value of x does $y(x) \to +\infty$?

Separable: $\frac{dy}{1+y^2} = \frac{dx}{x^2-4} \Rightarrow \arctan y = \frac{1}{4}\ln\left(\frac{2-x}{2+x}\right) + C$. Using $y(0) = 1$ gives $\arctan(1) = \frac{\pi}{4} = 0 + C \Rightarrow C = \frac{\pi}{4}$. Blow-up $y \to \infty$ occurs when $\arctan y \to \frac{\pi}{2}$, so $\frac{1}{4}\ln\left(\frac{2-x}{2+x}\right) + \frac{\pi}{4} = \frac{\pi}{2} \Rightarrow \ln\left(\frac{2-x}{2+x}\right) = \pi \Rightarrow \frac{2-x}{2+x} = e^{\pi}$. Solving for x.

20 **[Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

Solve $x^2 y' + xy = \frac{1}{x^2+1}$ for $x > 0$, with $y(1) = \frac{\pi}{4}$. What is $y(\sqrt{3})$?

Standard form: $y' + \frac{1}{x}y = \frac{1}{x(x^2+1)}$. $\mu = x$. Multiplying gives $(xy)' = \frac{1}{x(x^2+1)} = \frac{1}{x} - \frac{x}{x^2+1}$. Integrating gives $xy = \ln x - \frac{1}{2}\ln(x^2+1) + C$. Using $y(1) = \pi/4$ yields $\pi/4 = -\frac{1}{2}\ln 2 + C \Rightarrow C = \frac{\pi}{4} + \frac{1}{2}\ln 2$. At $x = \sqrt{3}$, $\sqrt{3}y = \frac{1}{2}\ln 3 - \frac{1}{2}\ln 4 + \frac{\pi}{4} + \frac{1}{2}\ln 2 = \frac{\pi}{4} + \frac{1}{2}\ln(3/2)$. Thus.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: General Solution Structure

What is the structure of the general solution to a linear nonhomogeneous equation $L[y] = g(x)$?

a. $y(x) = y_c(x) + y_p(x)$**, where** y_c **is the complementary solution and** y_p **is any particular solution** ✓

b. $y(x) = y_c(x) \cdot y_p(x)$

c. $y(x) = y_p(x)$ only

d. $y(x) = c_1 y_p(x) + c_2 y_c(x)$

By linearity, the general solution of a nonhomogeneous linear equation is the sum of the complementary solution y_c to the homogeneous equation and any particular solution y_p.

22 [Foundational] • Target: General Solution Structure

Find a particular solution y_p to $y'' - 5y' + 6y = 12$.

a. $y_p = 2$ ✓

b. $y_p = 12$

c. $y_p = 2x$

d. $y_p = 6$

Try constant $y_p = A$. Then $y_p' = 0, y_p'' = 0$. Substituting: $6A = 12 \Rightarrow A = 2$.

23 [Foundational] • Target: General Solution Structure

What is the trial particular solution y_p for $y'' + 4y = 3x^2 - 1$?

a. $y_p = Ax^2 + Bx + C$ ✓

b. $y_p = Ax^2$

c. $y_p = Ax^2 + B$

d. $y_p = Ax^3 + Bx^2$

For a quadratic polynomial forcing term $g(x) = 3x^2 - 1$, the standard trial solution is a complete quadratic polynomial $y_p = Ax^2 + Bx + C$.

24 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

Find a particular solution to $y'' + y = 4e^{2x}$.

a. $y_p = \frac{4}{5}e^{2x}$ ✓

b. $y_p = 4e^{2x}$

c. $y_p = 2e^{2x}$

d. $y_p = \frac{1}{2}e^{2x}$

Try $y_p = Ae^{2x}$. $y_p'' = 4Ae^{2x}$. Substitute: $4Ae^{2x} + Ae^{2x} = 5Ae^{2x} = 4e^{2x} \Rightarrow 5A = 4 \Rightarrow A = 4/5$.

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25 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

What is the correct undetermined coefficients form for $y'' + 9y = 5\cos(2x)$?

a. $y_p = A\cos(2x) + B\sin(2x)$ ✓

b. $y_p = A\cos(2x)$

c. $y_p = Ax\cos(2x)$

d. $y_p = A\sin(2x)$

Because $\cos(2x)$ produces both sines and cosines upon differentiation, the complete form is $A\cos(2x) + B\sin(2x)$.

26 [Foundational] • Target: Exponential and Sinusoidal Forcing Functions

Find y_p for $y'' + 9y = 5\cos(2x)$.

a. $y_p = \cos(2x)$ ✓

b. $y_p = 5\cos(2x)$

c. $y_p = \frac{5}{13}\cos(2x)$

d. $y_p = \sin(2x)$

Let $y_p = A\cos(2x)$. $y_p'' = -4A\cos(2x)$. $-4A\cos(2x) + 9A\cos(2x) = 5A\cos(2x) = 5\cos(2x) \Rightarrow 5A = 5 \Rightarrow A = 1$.

27 [Foundational] • Target: The Modification Rule (Resonance / Overlap)

When must a trial particular solution y_p be multiplied by x (or x^2)?

a. **When any term in the trial solution is already a solution of the associated homogeneous equation** ✓

b. Whenever the forcing function is a polynomial

c. When the coefficient of y' is negative

d. Only when the right-hand side is zero

If a term in the proposed y_p is part of the complementary solution y_c, substituting it will yield zero; multiplying by x restores linear independence.

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28 [Foundational] • Target: The Modification Rule (Resonance / Overlap)

For $y'' - 4y' + 4y = 3e^{2x}$, the homogeneous solution is $y_c = (c_1 + c_2x)e^{2x}$. What is the correct trial form for y_p?

a. $y_p = Ax^2e^{2x}$ ✓

b. $y_p = Ae^{2x}$

c. $y_p = Axe^{2x}$

d. $y_p = Ax^3e^{2x}$

Both e^{2x} and xe^{2x} are in y_c. Multiplying by x^2 produces $y_p = Ax^2e^{2x}$, which is linearly independent of y_c.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Modification Rule (Resonance / Overlap)

What physical phenomenon occurs when an undamped mechanical system is driven at its natural resonance frequency ($y'' + \omega_0^2 y = F_0\cos(\omega_0 t)$)?

a. **Resonance, where amplitude grows linearly with time ($y_p = Ct\sin(\omega_0 t)$)** ✓

b. Immediate decay to zero

c. Critical damping

d. Chaotic motion

Because the forcing frequency equals the natural frequency, the particular solution takes the form $t\sin(\omega_0 t)$, producing unbounded linearly growing oscillations (resonance).

30 [Intermediate] • Target: The Principle of Superposition

If y_{p1} solves $L[y] = g_1(x)$ and y_{p2} solves $L[y] = g_2(x)$, what solves $L[y] = 2g_1(x) - 3g_2(x)$?

a. $2y_{p1} - 3y_{p2}$ ✓

b. $y_{p1}y_{p2}$

c. $\frac{y_{p1}}{y_{p2}}$

d. $y_{p1} + y_{p2}$

By the linearity of differential operators, $L[2y_{p1} - 3y_{p2}] = 2L[y_{p1}] - 3L[y_{p2}] = 2g_1(x) - 3g_2(x)$.

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31 [Intermediate] • Target: The Principle of Superposition

Can the Method of Undetermined Coefficients be used to solve $y'' + y = \tan x$?

a. No, because derivatives of $\tan x$ produce infinitely many non-terminating forms; Variation of Parameters is required ✓

b. Yes, with trial solution $A\tan x + B\sec x$

c. Yes, with trial solution $A\sin x/\cos x$

d. Only if x is small

Undetermined coefficients applies only to functions with finite dimensional spaces of derivatives (polynomials, exponentials, sines/cosines). For $\tan x$, variation of parameters must be used.

32 [Intermediate] • Target: The Principle of Superposition

Find a particular solution to $y'' + 4y = 8x + 12$.

a. $y_p = 2x + 3$ ✓

b. $y_p = 2x$

c. $y_p = 4x + 3$

d. $y_p = 2x^2 + 3x$

Let $y_p = Ax + B$. $y_p'' = 0$. $4(Ax + B) = 4Ax + 4B = 8x + 12 \Rightarrow 4A = 8 \Rightarrow A = 2$ and $4B = 12 \Rightarrow B = 3$. Thus $y_p = 2x + 3$.

33 [Intermediate] • Target: General Solution Structure

Consider the second-order nonhomogeneous linear differential equation $y'' - 2y' + y = e^x\sin(x)$. Let $y_p(x)$ be a particular solution obtained using the method of undetermined coefficients. Determine the value of $y_p''(\pi) - 2y_p'(\pi) + y_p(\pi)$.

a. 0 ✓

b. e^{π}

c. $-\pi e^{\pi}$

d. $e^{\pi}\sin(\pi)$

The problem asks for the value of the left-hand side of the differential equation $y'' - 2y' + y = e^x\sin(x)$ when evaluated at a particular solution $y_p(x)$. By definition, a particular solution $y_p(x)$ must satisfy the nonhomogeneous differential equation for all x. Therefore, when we substitute $y_p(x)$ into the differential equation, the left-hand side must equal the right-hand side. Specifically, for any particular solution $y_p(x)$, it must be true that $y_p''(x) - 2y_p'(x) + y_p(x) = e^x\sin(x)$ for a.

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34 [Intermediate] • Target: Exponential and Sinusoidal Forcing Functions

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?

a. 2 ✓

b. $2x$

c. $2\sin(2x)$

d. $2\cos(2x)$

The given differential equation is $y'' + 4y = 8$. This is a nonhomogeneous second-order linear differential equation. We use the method of undetermined coefficients to find a particular solution, $y_p(x)$. The nonhomogeneous term is a constant, 8. We guess a particular solution of the form $y_p(x) = A$, where A is a constant. Then, $y_p'(x) = 0$ and $y_p''(x) = 0$. Substituting these into the differential equation: $0 + 4(A) = 8$. Solving for A, we get $4A = 8$, which means $A = 2$. Therefore, the particular solut.

35 [Intermediate] • Target: The Modification Rule (Resonance / Overlap)

Consider the nonhomogeneous linear second-order differential equation $y'' + y = 5$. If $y_p(x) = A$ is a particular solution, what is the value of A?

a. 1

b. **5** ✓

c. 1/5

d. 0

The given differential equation is $y'' + y = 5$. We are looking for a particular solution of the form $y_p(x) = A$, where A is a constant. To find the value of A, we substitute $y_p(x)$ and its derivatives into the differential equation. If $y_p(x) = A$, then the first derivative is $y_p'(x) = 0$ and the second derivative is $y_p''(x) = 0$. Substituting these into the differential equation $y'' + y = 5$, we get: $0 + A = 5$ Therefore, $A = 5$.

36 [Intermediate] • Target: The Principle of Superposition

Consider the second-order linear nonhomogeneous differential equation: $y''(x) - 3y'(x) + 2y(x) = 4$. Find the particular solution, $y_p(x)$, using the method of Undetermined Coefficients.

a. $y_p(x) = 2$ ✓

b. $y_p(x) = 2x$

c. $y_p(x) = 4$

d. $y_p(x) = 4x$

The given differential equation is $y''(x) - 3y'(x) + 2y(x) = 4$. Since the right-hand side is a constant, we guess a particular solution of the form $y_p(x) = A$, where A is a constant. Then, $y_p'(x) = 0$ and $y_p''(x) = 0$. Substituting these into the differential equation: $0 - 3(0) + 2(A) = 4$ $2A = 4$ $A = 2$ Therefore, the particular solution is $y_p(x) = 2$.

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37 [Intermediate] • Target: General Solution Structure

An undamped oscillator satisfies $y'' + 4y = 3\cos 2t$ (forcing at exactly the natural frequency), $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/4)$.

a. $-3\pi/16$

b. $3\pi/16$ ✓

c. $3\pi/8$

d. $\pi/4$

Homogeneous: $r^2 + 4 = 0 \Rightarrow y_c = C_1\cos 2t + C_2\sin 2t$. The forcing frequency matches the natural frequency exactly (resonance), so the trial needs t: $y_p = t(A\cos 2t + B\sin 2t)$. Substituting and matching gives $A = 0$, $B = \frac{3}{4}$, so $y_p = 34t2t$. With both conditions zero, $y_c \equiv 0$. So $y = 34t2t$, and $y(\pi/4) = \frac{3}{4} \cdot \frac{\pi}{4} \cdot \sin\frac{\pi}{2} = \frac{3\pi}{16}$ — the resonant amplitude $34t$ grows without bound as $t \to \infty$, unlike a non-resonant driven system.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Exponential and Sinusoidal Forcing Functions • Cross-Lesson: 1.01 Separable Equations

Find a particular solution of $y'' - y' - 6y = e^{2x} + 18$.

a. $14e^2x - 3$

b. $-14e^2x - 3$ ✓

c. $-14xe^2x - 3$

d. $-14e^2x + 3$

Split the forcing (the superposition note): for e^{2x}, $r = 2$ isn't a root of $r^2 - r - 6 = 0$ (roots $3, -2$), so trial Ae^{2x} gives $(4 - 2 - 6)A = 1 \Rightarrow -4A = 1 \Rightarrow A = -\frac{1}{4}$. For the constant 18, trial B gives $-6B = 18 \Rightarrow B = -3$. Adding the two pieces: $y_p = -14e^2x - 3$.

39 [Challenging] • Target: The Modification Rule (Resonance / Overlap) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Find a particular solution of $y'' - 4y' + 3y = 6x + 4$.

a. $2x - 4$

b. $2x + 2$

c. $2x + 4$ ✓

d. $4x + 4$

Homogeneous roots are $r = 1, 3$ (from $r^2 - 4r + 3 = 0$), and $r = 0$ isn't among them, so no resonance: trial $y_p = Ax + B$ gives $y_p' = A$, $y_p'' = 0$. Substituting: $-4A + 3(Ax + B) = 3Ax + (3B - 4A) = 6x + 4$. Matching: $3A = 6 \Rightarrow A = 2$; $3B - 4A = 4 \Rightarrow 3B = 4 + 8 = 12 \Rightarrow B = 4$. So $y_p = 2x + 4$.

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40 [Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' - y' - 2y = 3e^{2x}$, $y(0) = 1$, $y'(0) = 6$. Find $y(1)$.

a. $3e^2 + e^{-1}$

b. $e^2 - e^{-1}$

c. $3e^2 - e$

d. $3e^2 - e^{-1}$ ✓

Homogeneous: $r^2 - r - 2 = 0 \Rightarrow r = 2, -1$, $y_c = C_1e^{2x} + C_2e^{-x}$. $r = 2$ matches the forcing, so the trial needs x: $y_p = Axe^{2x}$. Substituting gives $3Ae^{2x} = 3e^{2x} \Rightarrow A = 1$, so $y_p = xe^{2x}$.
$y = C_1e^{2x} + C_2e^{-x} + xe^{2x}$. $y(0) = C_1 + C_2 = 1$; $y' = 2C_1e^{2x} - C_2e^{-x} + e^{2x} + 2xe^{2x}$,
$y'(0) = 2C_1 - C_2 + 1 = 6 \Rightarrow 2C_1 - C_2 = 5$; adding to the first, $3C_1 = 6 \Rightarrow C_1 = 2$, $C_2 = -1$. So
$y = 2e^{2x} - e^{-x} + xe^{2x} = (x + 2)e^{2x} - e^{-x}$, and $y(1) = 3e^2 - e^{-1}$.

41 [Challenging] • Target: General Solution Structure • Cross-Lesson: 5.01 Variation of Parameters

Solve $y'' + 9y = 2\sin 3t$, $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/3)$.

a. $\pi/9$ ✓

b. $-\pi/9$

c. $\pi/3$

d. $2\pi/9$

Homogeneous: $r^2 + 9 = 0 \Rightarrow r = \pm 3i$, $y_c = C_1\cos 3t + C_2\sin 3t$. The forcing $\sin 3t$ matches this pair exactly (resonance), so the trial needs t: $y_p = t(A\cos 3t + B\sin 3t)$. Substituting and matching gives $A = -\frac{1}{3}$, $B = 0$, so $y_p = -\frac{t}{3}\cos 3t$. With both conditions zero, $y_c \equiv 0$. So $y = -\frac{t}{3}\cos 3t$, and $y(\pi/3) = -\frac{\pi/3}{3}\cos\pi = -\frac{\pi}{9}(-1) = \frac{\pi}{9}$.

42 [Challenging] • Target: Exponential and Sinusoidal Forcing Functions • Cross-Lesson: 1.01 Separable Equations

Solve $y'' - 7y' + 12y = -56e^{-4x} - 144x$, $y(0) = -10$, $y'(0) = -15$. Find $y(1)$.

a. $e^{-4} + 19 + e^3 + e^4$

b. $-2e^4 - 2e^3 - 38 - \frac{2}{e^4}$

c. $-\frac{e^4}{2} - \frac{e^3}{2} - \frac{19}{2} - \frac{1}{2e^4}$

d. $-e^4 - e^3 - 19 - e^{-4}$ ✓

Homogeneous: $r^2 - 7r + 12 = 0$ has roots $r = 4, 3$, so $y_c = C_1e^{4x} + C_2e^{3x}$. Split the forcing (the superposition note): for $-56e^{-4x}$, -4 isn't a root, trial Ae^{-4x}; for $-144x$, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = -12x - 7 - e^{-4x}$. With $C_1 = -1$, $C_2 = -1$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

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43 [Challenging] • Target: The Modification Rule (Resonance / Overlap) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y'' + 3y' + 2y = 20e^{3x} - 4x$, $y(0) = -1$, $y'(0) = 8$. Find $y(1)$.

a. $-e^3 - 1 + \frac{2}{e^2} + \frac{3}{e}$

b. $-\frac{3}{e} - \frac{2}{e^2} + 1 + e^3$ ✓

c. $-\frac{6}{e} - \frac{4}{e^2} + 2 + 2e^3$

d. $-\frac{3}{2e} - e^{-2} + \frac{1}{2} + \frac{e^3}{2}$

Homogeneous: $r^2 + 3r + 2 = 0$ has roots $r = -1, -2$, so $y_c = C_1e^{-x} + C_2e^{-2x}$. Split the forcing (the superposition note): for $20e^{3x}$, 3 isn't a root, trial Ae^{3x}; for $-4x$, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = -2x + e^{3x} + 3$. With $C_1 = -3$, $C_2 = -2$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

44 [Challenging] • Target: The Principle of Superposition • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' + 4y' + 3y = -35e^{4x} - 9x$, $y(0) = 4$, $y'(0) = -14$. Find $y(1)$.

a. $-e^4 - \frac{2}{e} + \frac{3}{e^3} + 1$ ✓

b. $-1 - \frac{3}{e^3} + \frac{2}{e} + e^4$

c. $-2e^4 - \frac{4}{e} + \frac{6}{e^3} + 2$

d. $-\frac{e^4}{2} - e^{-1} + \frac{3}{2e^3} + \frac{1}{2}$

Homogeneous: $r^2 + 4r + 3 = 0$ has roots $r = -1, -3$, so $y_c = C_1e^{-x} + C_2e^{-3x}$. Split the forcing (the superposition note): for $-35e^{4x}$, 4 isn't a root, trial Ae^{4x}; for $-9x$, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = -3x - e^{4x} + 4$. With $C_1 = -2$, $C_2 = 3$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

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45 [Challenging] • Target: General Solution Structure • Cross-Lesson: 5.01 Variation of Parameters

Solve $y'' - y' - 2y = 10e^{-3x} + 4x$, $y(0) = 3$, $y'(0) = -12$. Find $y(1)$.

a. $-2e^2 - 1 + e^{-3} + \frac{3}{e}$ ✓

b. $-\frac{3}{e} - e^{-3} + 1 + 2e^2$

c. $-4e^2 - 2 + \frac{2}{e^3} + \frac{6}{e}$

d. $-e^2 - \frac{1}{2} + \frac{1}{2e^3} + \frac{3}{2e}$

Homogeneous: $r^2 - r - 2 = 0$ has roots $r = 2, -1$, so $y_c = C_1e^{2x} + C_2e^{-x}$. Split the forcing (the superposition note): for $10e^{-3x}$, -3 isn't a root, trial Ae^{-3x}; for $4x$, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = -2x + 1 + e^{-3x}$. With $C_1 = -2$, $C_2 = 3$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

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